EX:1	SETTING UP PYTHON ENVIRONMENT AND LIBRARIES – JUPYTER NOTEBOOK
DATE:	

AIM:

To install Python, configure Jupyter Notebook, understand the notebook environment, execute Python programs, create Markdown cells, and use basic Jupyter Notebook features for Data Analytics applications

ALGORITHM:

- Step 1: Download the latest Python installer from the official Python website.
- Step 2: Run the installer, select "Add Python to PATH", and click Install Now.
- Step 3: Verify the Python installation by executing the command python --version in the Command Prompt.
- Step 4: Open the Command Prompt and install Jupyter Notebook using the command pip install notebook.
- Step 5: Launch Jupyter Notebook by executing the command jupyter notebook.
- Step 6: Wait for the browser to automatically open the Jupyter Notebook Dashboard.
- Step 7: Click New → Python 3 (ipykernel) to create a new notebook.
- Step 8: Rename the notebook with an appropriate title.
- Step 9: Create Code cells and Markdown cells as required.
- Step 10: Write and execute the program using Shift + Enter.
- Step 11: Save the notebook and verify that it executes successfully.
- Step 12: End the program

PROGRAM:

```python
print("Welcome to Data Analytics Lab")
```

```python
a = 20
b = 10

print("Addition =", a+b)
print("Subtraction =", a-b)
print("Multiplication =", a*b)
print("Division =", a/b)
```

```python
name = "John"
dept = "AI&DS"
cgpa = 8.9

print("Name:", name)
print("Department:", dept)
print("CGPA:", cgpa)
```

```python
import numpy as np

a = np.array([10,20,30,40,50])

print(a)
print("Mean =", np.mean(a))
print("Sum =", np.sum(a))
```

```python
import pandas as pd

data = {
    'Name':['John','David','Alex'],
    'Marks':[85,90,95]
}

df = pd.DataFrame(data)

print(df)
```

```python
import matplotlib.pyplot as plt

x = [1,2,3,4,5]
y = [10,20,15,25,30]

plt.plot(x,y)
plt.title("Sample Line Graph")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
df = pd.read_csv(r"C:\Users\HDC0422048\Downloads\archive (4)\malnutrition_data (1).csv")
le = LabelEncoder()
df["nutrition_status"] = le.fit_transform(df["nutrition_status"])
X = df.drop("nutrition_status", axis=1)
y = df["nutrition_status"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy * 100)
print(classification_report(y_test, y_pred))
```

OUTPUT:

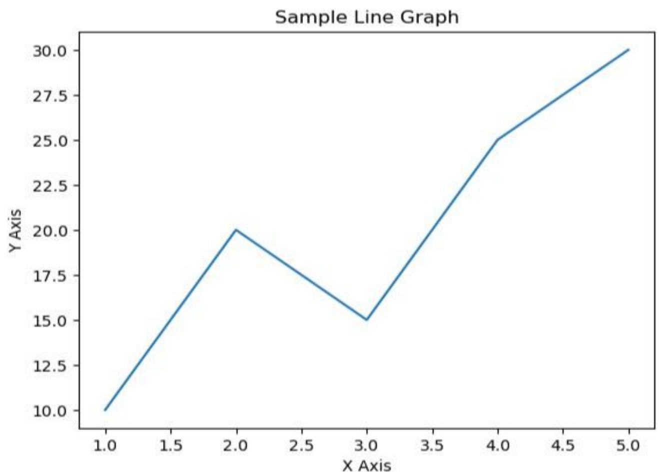
```
Welcome to Data Analytics Lab
```

```
Addition = 30
Subtraction = 10
Multiplication = 200
Division = 2.0
```

```
Name: John
Department: AI&DS
CGPA: 8.9
```

```
[10 20 30 40 50]
Mean = 30.0
Sum = 150
```

```
    Name  Marks
0   John     85
1  David     90
2   Alex     95
```




```
Accuracy: 95.3
              precision    recall  f1-score   support

           0       0.87      0.89      0.88       198
           1       0.99      0.99      0.99       734
           2       0.79      0.74      0.76        68

    accuracy                           0.95      1000
   macro avg       0.88      0.87      0.88      1000
weighted avg       0.95      0.95      0.95      1000
```

RESULT:

Python and Jupyter Notebook were installed successfully. A new Jupyter Notebook was created, renamed, and executed without errors. Code and Markdown cells were added and run successfully. Thus, the Python programming environment was set up successfully.